THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** systematic investigation	Core idea of the chapter
2	**(c)** calendar based on sky observations	Calendars link to Sun and Moon
3	**(d)** microscope	Magnifies tiny objects
4	**(c)** vitamin C	A deficiency disease
5	**(c)** take antibiotics only when prescribed	Complete the right dose
6	**(c)** movement of liquid iron in the core	Creates electric currents
7	**(d)** steel manufacturing	Melts and recycles scrap steel
8	**(c)** rough surface	More irregularities
9	**(a)** smaller area gives more pressure	P = F/A
10	**(c)** storm surge	3–12 m high
11	**(c)** water vapour (gas)	At 100 °C at atmospheric pressure
12	**(d)** fixed ratio	Fixed composition
13	**(d)** carbon dioxide and water	Both contain only non-metal elements
14	**(a)** adding more solvent	More capacity
15	**(c)** inverted	Depends on distance
16	**(c)** decreases	Waning
17	**(b)** frog	Eats a herbivore
18	**(c)** terrestrial ecosystem	Land-based
19	**(d)** a strong greenhouse effect	Dense CO_2 atmosphere
20	**(c)** three oxygen atoms	O_3

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It converts milk into curd (1). It is also used in fermentation in food industry (1).	1+1
22	The natural ability of the body to fight diseases (1). It is carried out by the immune system (1).	1+1
23	Oersted (1). In 1820 (1).	1+1
24	Air is pushed out, reducing pressure inside (1). Outside air pressure is higher and presses it (1).	1+1
25	Bring a magnet near it (1). Iron sticks, sulfur remains (1).	1+1
26	kg/m³ (1). g/cm³ (1).	1+1
27	Positions of organisms in a food chain (1). Producers occupy the first level (1).	1+1
28	Right distance from the Sun (1). Atmosphere with oxygen and ozone (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Living things depend on air, water, sunlight and each other (1½). Understanding them helps protect life (1½).	1½+1½
30	Diagram (1). Spring and hook (1). Scale in newton and gram (1).	1+1+1
31	Particles need enough energy to escape (1½). At room temperature only slow evaporation occurs (1½).	1½+1½
32	Mirror reflects; lens refracts (1). Mirror converges; lens diverges (1). Mirror can form real images; lens forms virtual images (1).	1+1+1
33	Lunar year = 12 × 29.5 = 354 days (1). Gap = 11 days (1). About 3 years give ~33 days, one extra month (1).	1+1+1
34	More UV rays reach Earth (1½). They damage cells and ecosystems (1½).	1½+1½
35	Zinc–copper, aluminium–copper, iron–copper (any three) (2). Copper acts as positive (1).	2+1

	OR: Wind insulated wire around an iron nail (1). Connect ends to a cell (1). It picks up pins or clips while current flows (1).	1+1+1
36	Pollination of crops (1). Natural pest control (1). Fertile soil from decomposers (1).	1+1+1
	OR: Arrows show flow of energy and matter (1½). Energy passes from prey to predator (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Ayurveda: balance of body, mind and surroundings (1½). Dinacharya and ritucharya routines (1). Variolation against smallpox (1). Modern vaccines are safer and tested (1½).	1½+1+1+1½
	OR: Avoids pain (1). Saves cost (1). Prevents spread (1). Vaccines and hygiene (1). Healthy lifestyle reduces NCDs (1).	1+1+1+1+1
38	0.1 m²: 400 N/m² (1½). 0.05 m²: 800 N/m² (1½). Smaller area means more pressure (2).	1½+1½+2
	OR: Pressure differences cause wind (1). Sea and land breezes (1). Storms from low pressure over heated land (1). Cyclones from very low pressure over ocean (1). Speed larger with greater difference (1).	1+1+1+1+1
39	Solubility decreases (1). Excess solute separates (1). Crystals form (1). Example – sugar or salt crystals (1). Slow cooling gives bigger crystals (1).	1+1+1+1+1
	OR: A = 1 g/mL (1½). B = 0.8 g/mL (1½). C = 1.2 g/mL (1). B floats (1).	1½+1½+1+1
40	Web diagram with arrows (1). Producer identified (1). Consumers levels named (1). Removal increases mice and frogs (1). Hawks lose a food source and balance changes (1).	1+1+1+1+1
	OR: Vrikshayurveda text (1). Organic manure (1). Kunapa Jala (1). Chemical farming harms soil (1). Balanced approach (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER